

Final Report: Twitter Named Entity Recognition

BiLSTM, GloVe, Data Imbalance Handling, Focal Loss, DistilBERT, BERT-base and Hyperparameter Tuning

Prepared from the attached source document: `Final_Twitter_NER_Recognition.pdf`

Metric	Value
Training samples	2,394 sentences
Testing samples	3,850 sentences
Total tokens	108,375
Unique BIO tags	22
Vocabulary size	25,384
Chosen max padding length	29 tokens
Best reported model	Tuned BERT-base
Best reported micro F1	0.4418

End-to-End NER Pipeline



Executive Summary

This report consolidates the Twitter NER experiments into a polished final document. The project begins with a BiLSTM baseline using GloVe embeddings, then improves imbalance handling with augmentation, class weights and focal loss, and finally transitions to transformer-based token classification. The best result is obtained by the tuned BERT-base model with an entity-level micro F1 of 0.4418.

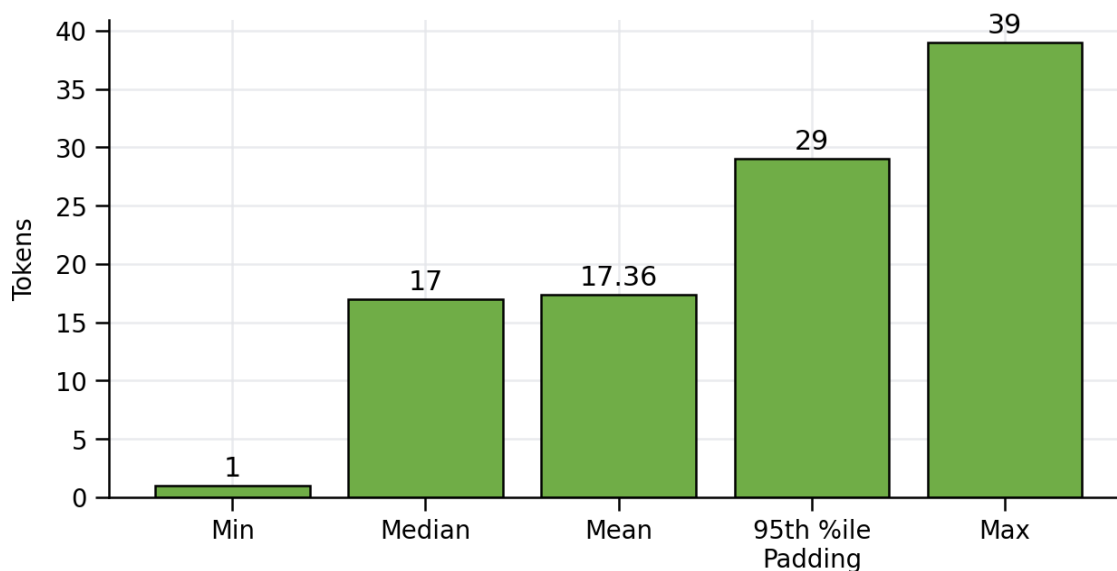
1. Dataset Overview

The dataset follows CoNLL-style BIO tagging where each token is paired with an entity tag and blank lines separate sentences. Twitter NER is challenging because text is noisy, informal, short, and highly imbalanced toward the non-entity 'O' class.

Dataset Split: Training vs Testing Sentences



Sentence Length Summary



- 6,244 total sentences and 108,375 tokens were processed.
- The 95th percentile sentence length produced a max padding length of 29.
- The vocabulary contained 25,384 words including and .
- The tag schema contained 22 labels including BIO entity labels and the null/padding marker.

2. Modeling Approach

NER Model Architecture

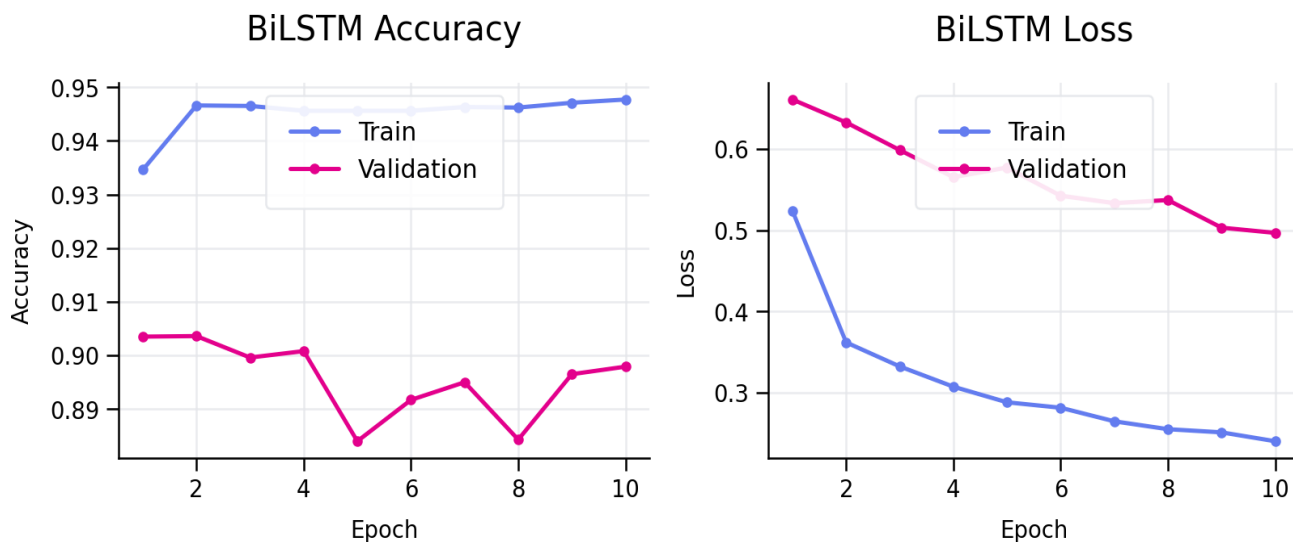


Baseline: BiLSTM + GloVe | Advanced: BERT-base / DistilBERT token classification

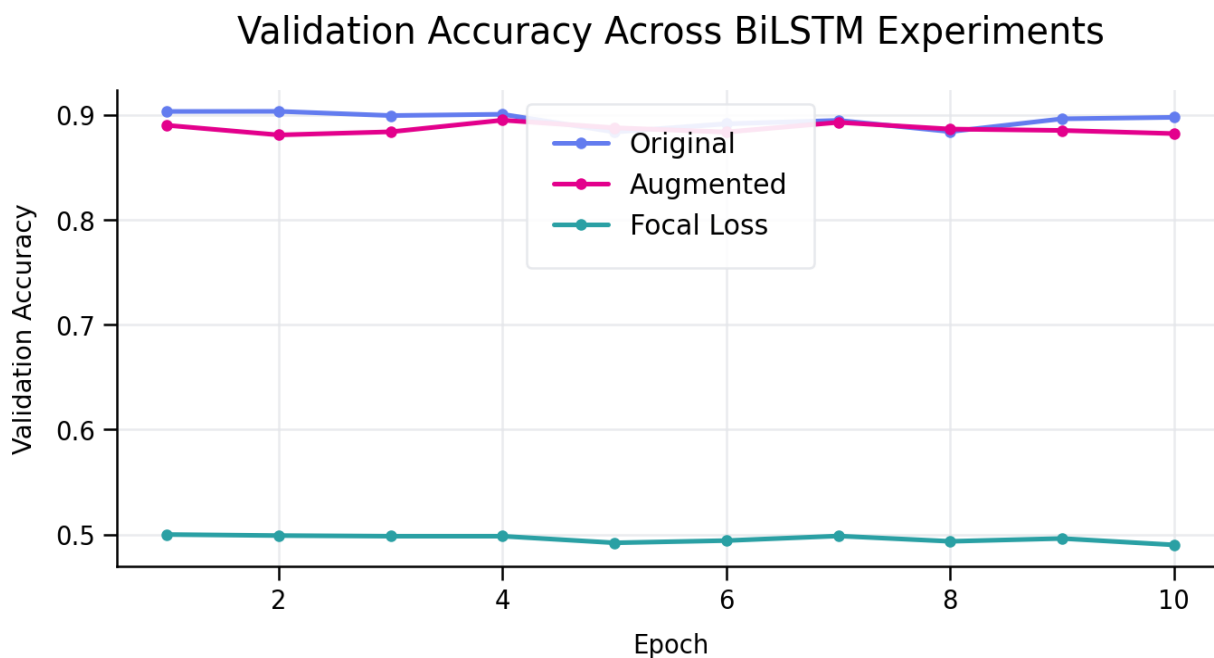
The baseline model used a non-trainable 100-dimensional GloVe embedding matrix, followed by a Bidirectional LSTM and a TimeDistributed softmax classifier. Transformer experiments used Hugging Face token classification models with subword label alignment and SeqEval evaluation.

Component	Configuration
Embedding	GloVe 100d; embedding matrix shape (25,384, 100); frozen
Encoder	Bidirectional LSTM, 128 units, return_sequences=True
Classifier	TimeDistributed Dense, 22 output tags, softmax
Parameters	2,778,550 total; 240,150 trainable
Transformer path	DistilBERT attempted; BERT-base trained and tuned

3. Training Diagnostics and Imbalance Handling



The original BiLSTM reached 0.9477 training accuracy and 0.8979 validation accuracy, but entity-level F1 remained low because the majority 'O' label dominated token-level accuracy.



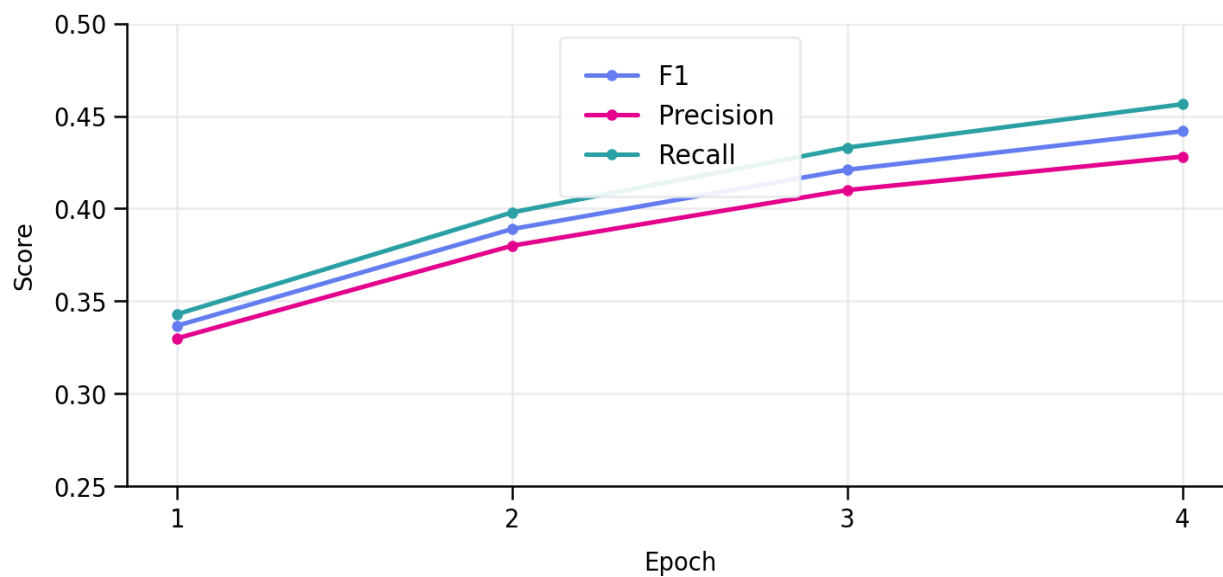
Strategy	Purpose	Observed effect
Synonym augmentation	Increase entity-bearing samples	Generated 367 augmented samples; total training set 5,155
Class weights	Higher penalty for rare tags	Improved recall but increased false positives
Focal loss	Emphasize difficult/minority examples	Improved BiLSTM entity-level F1 to 0.12
SeqEval	Entity-level evaluation	More reliable than token accuracy for NER

4. Transformer Results

DistilBERT was prepared with tokenizer, token-classification head, aligned labels and PyTorch datasets, but the training run hit a tensor shape mismatch. BERT-base was then reimplemented with dynamic padding and trained successfully through the Hugging Face Trainer API.

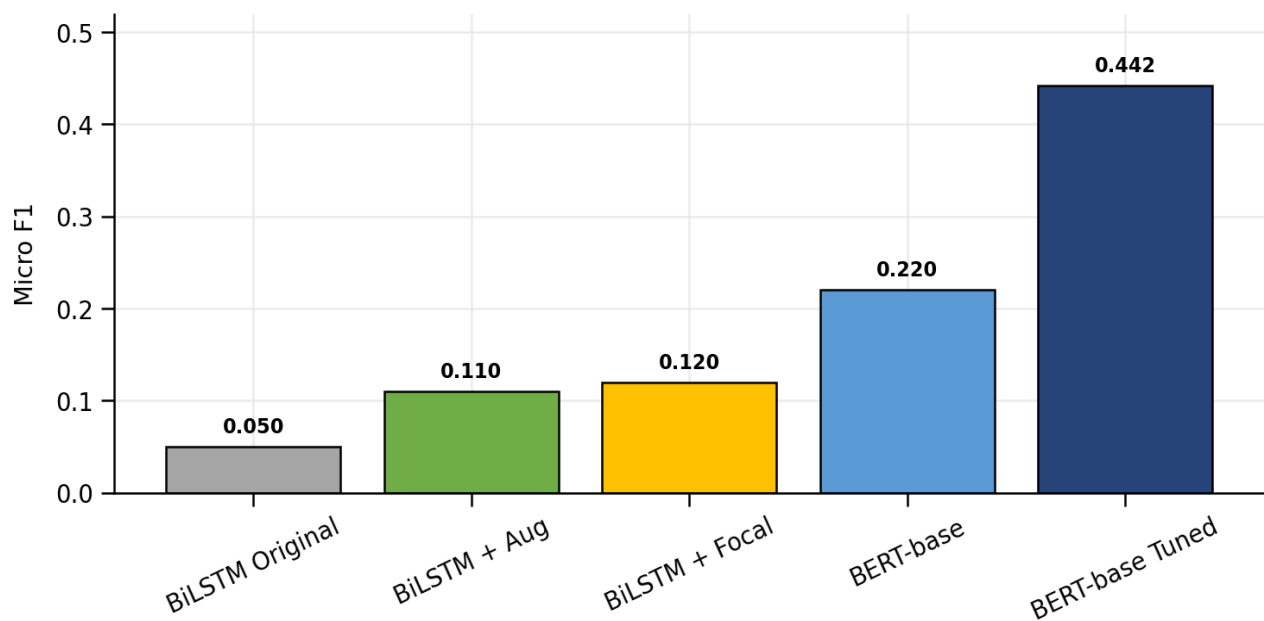
Model	Configuration	Result
DistilBERT	distilbert-base-uncased; aligned BIO labels	Training blocked by input shape mismatch
BERT-base	bert-base-uncased; 1 epoch	Precision 0.2617, Recall 0.1918, F1 0.2213
BERT-base tuned	Optuna: lr≈2.89e-5, epochs=4, batch=8	Best F1 0.4418

Tuned BERT-base Entity Metrics



5. Final Model Comparison

Model Comparison by Entity-level Micro F1



Rank	Model	Micro F1	Interpretation
1	BERT-base Tuned	0.4418	Best overall
2	BERT-base	0.2200	Strong transformer baseline
3	BiLSTM + Focal	0.1200	BiLSTM experiment
4	BiLSTM + Aug	0.1100	BiLSTM experiment
5	BiLSTM Original	0.0500	BiLSTM experiment

6. Sample Inference and Recommendations

Sentence	Representative outcome
Google's CEO Sundar Pichai visited the White House today.	BERT-base tagged Google/Sundar/Pichai-like tokens as person; outputs still need cleaning
Apple released a new iPhone model in California.	California was detected as geo-location.
The game between Fnatic and G2 Esports will be in Berlin.	Berlin was detected as geo-location; sports-team/product recognition remained difficult

- Use tuned BERT-base as the strongest current baseline.
- Fix DistilBERT shape mismatch by using DataCollatorForTokenClassification or enforcing [batch_size, sequence_length] tensors.
- Move to BERTweet for tweet-specific language, hashtags, mentions and slang.
- Use entity-level SeqEval F1 as the primary metric; use token accuracy only as secondary.
- Package the project into enterprise-style pipelines with configs, logging, exceptions, timers and Optuna tuning.

Conclusion

The project demonstrates a clear progression from a traditional BiLSTM baseline to transformer-based NER. The tuned BERT-base result is the strongest outcome and provides a practical foundation for future DistilBERT/BERTweet production pipelines.